PURNIKA NAGA DURGA JYOTHI

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SUMMARY

Computer Science and Engineering undergraduate with hands-on experience in Python programming, backend development, AI applications, and machine learning. Familiar with FastAPI, REST APIs, databases, and data processing using Python libraries. Interested in applying problem-solving skills to develop practical Python-based solutions and learn new technologies.

TECHNICAL SKILLS

Programming: Python, Java, SQL, JavaScript, C++

Python Frameworks and Libraries: FastAPI, Pydantic, TensorFlow, Scikit-learn, OpenCV, BeautifulSoup, Selenium

Backend: REST APIs, JWT Authentication, Node.js, Express.js

Databases: PostgreSQL, MySQL, MongoDB, Redis

AI/ML: LLMs, RAG, GenAI, NLP, Computer Vision

Core CS: Data Structures and Algorithms, OOP, DBMS

Tools: Git, GitHub, Postman, Docker

WORK EXPERIENCE

AI Engineer Intern (Project Lead) – Springer Capital |Remote| Aug 2026-Present

Capture the Flag Intern Challenge (AI Track) - Compliance Document Review App

- Led the Generative AI track, building a Python FastAPI service that integrates 3 LLM providers (Gemini, Groq, OpenRouter) for automated document summarization and compliance analysis.
- Designed a 5-component service architecture spanning Next.js, Express/TypeScript, FastAPI, PostgreSQL, and AI services, connected through REST APIs and JWT authentication.
- Built a dedicated PII protection layer covering 4 sensitive data categories, converting detected information into stable placeholders so only zero-PII content reaches the AI engine.
- Implemented 2 AI-service execution paths (Gemini / Mock AI) with automated fallback across 3 LLM providers, generating compliance flags, suggestions, and risk scores while handling API rate limits and provider failures.

PROJECTS

LLM Data Analyst Agent with Guardrails

Python, FastAPI, LangGraph, PostgreSQL, Pydantic, Docker, Gemini/Groq

• Built a Python agent capable of planning and executing multi-step tasks, answering natural-language questions about PostgreSQL databases using 3
• Implemented 3 security measures—input sanitization, read-only SQL allowlisting, and query validation—to prevent destructive and unauthorized database operations.
• Enforced structured outputs using Pydantic schemas and automatic retries on validation failures, reducing malformed responses by 30%.
• Implemented per-request token and cost tracking, with automatic fallback across 3 LLM providers to handle rate limits and service outages.
• Integrated PII masking to replace sensitive information with placeholders before LLM processing; evaluated performance across 100 sample queries, achieving 90% answer accuracy.

Multimodal Deepfake Detection System

Python, TensorFlow, OpenCV, ResNet-50, CNN - Published at IEEE ICSIST 2026

• Engineered a multimodal deep learning pipeline in Python and TensorFlow for AI-generated image detection, achieving **90% test accuracy**.
• Combined RGB, Fast Fourier Transform (FFT) frequency-domain analysis, and noise residual features to detect image-level inconsistencies; applied Grad-CAM for model explainability.
• Evaluated performance using precision, recall, F1-score, and confusion matrix; published findings as a peer-reviewed conference paper.

Smart Queue Management System

React.js, Spring Boot, PostgreSQL, Redis, WebSockets, REST APIs

• Developed a full-stack queue management system with 5+ REST API operations for queue creation, token management, user handling, and service-counter operations
• Designed a layered backend (Controller, Service, Repository) with PostgreSQL data models for structured queue and user management.
• Implemented WebSocket-based real-time updates to synchronize queue positions and service status across users, with Redis caching for frequently accessed queue data.

EDUCATION

SRM Institute of Science and Technology **2023 – 2027**
Bachelor's Degree in Computer Science and Engineering - CGPA: 8.28/10

PUBLICATIONS

• **Multimodal Deepfake Detection Using Deep Learning Framework** - Presented at the International Conference on Secure Information Systems and Technologies (ICSIST) 2026, IEEE. IEEE Xplore

CERTIFICATIONS

• Generative AI – Microsoft
• Machine Learning, JAVA – NPTEL,
• CSS3 - Infosys

ACHIEVEMENTS & LEADERSHIP

• **Kaggle:** Earned Kaggle Expert tier with 3 public ML notebooks
• **Leadership:** Conducted a Python workshop for 60+ students at SRM IST on APIs and data handling
• **Hackathon:** Top 10 finalist at the AI Agent Hackathon among 60+ teams built a multi-tool AI agent that answers business questions over a database, using Python, LangGraph, and FastAPI